

Course: Natural Language Processing

PyTorch Tutorial and Assignment 2/3

AIMS CAMEROON / 2024–2025

Shamsuddeen Hassan Muhammad, PhD
Imperial College London

Date: March 27, 2025

Neural Machine Translation with Attention

Sequence-to-sequence models with attention have become a key component of modern machine translation systems.

Click here for the PyTorch Tutorial

Translation with a Sequence to Sequence Network and Attention (PyTorch Official Tutorial)

Assignment

After completing the tutorial above, apply your understanding with the following task:

Part 1: Follow the Tutorial

- Carefully go through the PyTorch tutorial on sequence-to-sequence translation with attention.
- Implement the code provided and understand each component.

Part 2: Translation on a New Dataset

- Choose a different dataset for machine translation (you are encouraged to use African language datasets).
- Suggested resource: MAFAND - African Machine Translation Datasets
- Train your own model using the new dataset.
- Evaluate the translation results and describe your findings.